00-P-636D
February 2, 1977
SUPERSEDING
00-P-636C
February 5, 1974

FEDERAL SPECIFICATION

PRESSES, ARBOR, HAND-OPERATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand operated mechanical presses and hand operated hydraulic presses.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, styles and sizes. This specification covers hand operated arbor presses of the following types, classes, and styles and the sizes shown in tables I through V. The type, class, style and size press required shall be as specified (see 6.2.1).

Type I - Bench or pedestal mounted
  Class 1 - Mechanical
  Class 2 - Hydraulic

Type II - Column or wall mounted, mechanical

Type III - Floor mounted
  Class 1 - Mechanical
    Style A - Elevating table
    Style B - Swinging table

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Class 2 - Hydraulic
Style A - Standard (non-rolling) table
Style B - Roller mounted table

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for Bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, Mo., Forth Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-M-18058 - Machinery, Metal and Woodworking, Preparation for Delivery of

Military Standard:

MIL-STD-130 - Identification Marking of U.S. Military Property

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Occupational Safety and Health Act (OSHA) of 1970:

29 CFR 1910 - Occupational Safety and Health Administration Standards.

National Bureau of Standards (NBS) Handbook:

H-28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Gear Manufacturers Association (AGMA):

AGMA 360 - Manual for Machine Tool Gearing.

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Ave., NW, Washington, DC 20005.)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the presses are classified herein as follows:

Requirement	Paragraph
First article approval	3.2
General requirements (all types, classes, styles, and sizes)	3.3
Detailed requirements	3.4
Type I, Class 1	3.4.1
Type I, Class 2	3.4.2

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Requirement	Paragraph
Type II	3.4.3
Type III, Class 1	3.4.4
Type III, Class 2	3.4.5
Performance requirements (all types, classes, styles and sizes)	3.5

3.2 First article. When specified (see 6.2.1), the contractor shall furnish one complete press for first article inspection and approval (see 4.2 and 6.4).

3.3 General requirements.

3.3.1 Design. The press shall be new and one of the manufacturer's current models capable of performing pressing operations in accordance with the requirements herein. The press shall include all components, parts and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement and repair.

3.3.2 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.3.3 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the press presenting safety hazards which can be guarded without substantial interference to operation of the machine. All guards shall provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the press shall comply with the standards promulgated under OSHA applicable to the press manufacturer. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

3.3.4 Lubrication. Means shall be provided to insure adequate lubrication to all moving parts. All oil holes, grease fittings and filler caps shall be easily accessible.

3.3.5 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensional and tolerance standards which will permit installation of replacement parts without modification to the part or equipment.

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3.3.6 Construction. The press shall be constructed of parts which are new, without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the press to 110 percent of its maximum rating and capacity without permanent distortion.

3.3.7 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging or filling with paste shall be used for reclaiming any defective part.

3.3.8 Welding, brazing or soldering. Welding, brazing or soldering shall be used only where specified in the original design. None of these operation shall be employed as a repair measure for any defective part.

3.3.9 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.10 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.11 Painting. Unless otherwise specified (see 6.2.1), the press shall be painted in accordance with the manufacturer's commercial practice.

3.3.12 Threads. All threaded parts shall be in the inch system and in accordance with H28.

3.3.13 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.14 Gears. All gears used in the press and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

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3.3.15 Fungus control. When required (see 6.2.1) fungus proofing shall be as specified.

3.3.16 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant nameplate, marked in accordance with MIL-STD-130, shall be securely attached to each press. The nameplate shall contain the information listed below. If the press is a special model, the model designation shall include the model of the basic standard press and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer serial number
Rated press tonnage
Date of manufacture
Contract Number or Order Number
Federal Stock Number or Plant Equipment Code

3.3.17 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each press. The following information shall be furnished on the chart or plate:

Points of application
Service interval
Type of lubricant
Viscosity
Military or Federal Specification

If there are no points on the press requiring lubrication, the chart or plate shall be inscribed "No lubrication required" in lieu of the information required above.

3.3.18 Technical data. Technical data shall be furnished as specified (see 6.3).

3.3.19 Workmanship. Workmanship of the press and accessories shall be commensurate with the requirements specified herein.

3.4 Detailed requirements.

3.4.1 Type I, class 1 presses.

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3.4.1.1 Frame. The type 1, class 1 press shall have a "C" or gap type frame. The frame shall support the ram, ram operating mechanism, and work support plate. The ram shall have accuracy to maintain the ram axis perpendicular to the table surface within 0.010 inch per 6 inches of ram length. The frame table shall have a slot aligned with the axis of the ram that will allow passage of an arbor of the largest diameter specified in table I. The frame throat depth shall accomodate workpieces of the largest diameter specified in table I. The table shall have means for mounting the work support plate (see 3.4.1.2). The table surface supporting the plate shall be finish machined so that the plate will rest squarely on the table surface without rocking. The frame of type I presses shall be designed for mounting onto a bench, workstand or pedestal. When specified (see 6.2.1), a pedestal or floor stand shall be supplied.

3.4.1.2 Work support plate. All type I presses shall have a circular work support plate having not less than four equally spaced slots. The smallest slot shall be sized to allow passage of an arbor having a diameter 20 to 25 percent of the largest diameter arbor size specified in table I. The other slots shall progressively increase in size, and the largest slot shall be sized to allow passage of an arbor having a diameter 80 to 90 percent of the largest diameter arbor size specified in table I. The plate shall be removable to allow the table slot (see 3.4.1.1) to be used when arbors of the largest diameter specified in table I are to be pressed. The plate shall rotate so that the desired plate slot can be readily positioned in alignment with the ram axis.

3.4.1.3 Ram and ram operating mechanism. The ram shall be extended toward the table by means of a hand lever operated pinion that meshes with a rack machined onto the surface of the ram. Presses having a tonnage rating of 3 tons, or less, may have a non-ratcheting type hand lever. Presses having a tonnage rating more than 3 tons shall have a ratcheting type hand lever and a hand wheel that allows the ram to be rapidly extended to contact the arbor being pressed, and allows the ram to be rapidly retracted after the pressing operation. Ratcheting type hand levers shall have a counterweight, or spring powered mechanism, that raises the lever to the uppermost position when hand force is released from the lever. The hand lever may be either of the single lever or compound lever type provided that the leverage ratio (ratio of pounds of ram force to pounds of hand force on the end of the operating lever) conforms to the requirements of table I. Compound lever type handles shall have means for rapidly changing to single-lever type action by moving or sliding a control.

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3.4.1.4 Sizes and capacities. The size and capacities of type I, class 1 presses shall conform to the requirements of table I.

3.4.2 Type I, class 2 presses.

3.4.2.1 Frame. Unless otherwise specified, the type I, class 2 press shall be of either the "C" frame or "4" post frame design. When specified (see 6.2.1), the type I, class 2 press shall be of the "C" frame design. The frame shall support the ram cylinder and pump assembly. The ram and cylinder axis shall be perpendicular to the work supporting surface of the table within 0.010 inch per 6 inches of ram length. The table structure shall have an opening aligned with the ram axis that will allow passage of an arbor of the largest diameter specified in table II. The worktable shall be designed to accommodate work support devices conforming to 3.4.2.4. The frame configuration shall enable the press to accommodate workpieces up to the largest workpiece diameter specified in table II. The distance between the ram nose and the worktable shall conform to the requirements of table II. The frame shall be designed for mounting onto a bench, workstand or pedestal. When specified (see 6.2.1), a pedestal or floor stand shall be furnished.

3.4.2.2 Hydraulic system. The press shall have a hydraulic system consisting essentially of a fluid reservoir, a pump, a cylinder and ram, a pressure release valve and a pressure gauge. The reservoir shall have a capacity not less than 1-1/4 times the volume of fluid required to fill the system and force the ram to the bottom of its stroke. The design of the pump, cylinder and ram shall enable the press to develop its rated tonnage as required by table II, without requiring more than 110 pounds of force applied to the pump lever. The hydraulic system shall have a pressure release valve that, when manually actuated, will release hydraulic pressure back to the fluid reservoir and allow the ram to automatically retract to the top of stroke position. Means shall be provided to prevent extension of the ram beyond its normal "end-of-stroke" position. The press shall have a pressure gauge calibrated in "tons of ram force" that enables the operator to monitor the ram force being exerted.

3.4.2.3 Ram. The press shall have means for rapidly extending the ram nose to contact the arbor prior to the arbor pressing operation. The ram extension method shall be either by means of a dual-stage hydraulic pump having a high volume-low pressure stage for rapid ram extension, or by means of a threaded ram extension that can be manually screwed down until it contacts the arbor being pressed. The ram piston travel shall conform to the requirements of table III.

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Table I. Sizes and capacities.

Type I - Class 2

Bench or Pedestal Mounted, Mechanical Presses

Size	Rated tonnage	Largest diameter workpiece accommodated, inches	Largest diameter arbor accommodated, inches	Distance, ram to table surface (ram up), inches	Leverage ratio
1/2	1/2	6-1/2	1	4-1/2	20:1
1	1	7	1	5	25:1
1-1/2	1-1/2	9	1	7-3/4	32:1
2	2	10	1-1/4	8	35:1
3A	3	12	1-5/8	11-1/4	36:1
3B	3	12	2	11-1/2	45:1
3C	3	12	2	18	45:1
3D	3	14	2-1/4	22	45:1
4A	4	15	2-1/2	12-1/2	60:1
4B	4	18	3-5/8	17-1/2	60:1
4C	4	15	2-1/2	18	60:1
4D	4	25	3	12	60:1
4E	4	15	2-1/2	23	60:1
5A	5	18	3-1/2	16	55:1
5B	5	25	3	12	70:1
5C	5	22	5	21	70:1
6A	6	15	2-1/2	15	66:1
6B	6	15-1/2	3	21	66:1
6C	6	15-1/2	5	15	66:1
8	8	18	3-5/8	17-1/2	100:1
10	10	25	3	12	120:1
12A	12	20	5	19	112:1
12B	12	22	3	17	150:1

Requirements are not less than those shown.

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3.4.2.4 Work support devices. A work support plate and a pair of matched V-blocks shall be furnished with each press. The work support plate shall conform to 3.4.1.2 (substituting "table II" wherever "table I" is mentioned in 3.4.1.2). The matched V-blocks shall be capable of supporting a round bar, or shaft, having a diameter of 1-inch or more, for shaft straightening operations.

3.4.2.5 Sizes and capacities. The sizes and capacities of type I, class 2 presses shall conform to the requirements of table II.

Table II. Sizes and capacities.

Type I - Class 2

Bench or Pedestal Mounted, Hydraulic Presses

Size	Rated tonnage	Largest diameter workpiece accommodated inches	Largest diameter arbor accommodated inches	Distance ram to table surface, (ram up), inches	Ram piston travel inches	Maximum operating force on pump lever pounds
10	10	17	1-1/4	9-1/2	6	110

Requirements are not less than those shown.

3.4.3 Type II, presses.

3.4.3.1 Frame. The press shall have a frame conforming to all requirements of 3.4.1.1 (substituting "table III" wherever "table I" is mentioned in 3.4.1.1), except that the frame shall be designed for column or wall mounting in lieu of bench, workstand or pedestal mounting.

3.4.3.2 Work support plate. The press shall have a work support plate conforming to all requirements of 3.4.1.2 (substituting "table III" wherever "table I" is mentioned in 3.4.1.2).

3.4.3.3 Ram and ram operating mechanism. The press shall have a ram and ram operating mechanism conforming to all requirements of 3.4.1.3 substituting "table III" wherever "table I" is mentioned in 3.4.1.3).

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3.4.3.4 Sizes and capacities. The sizes and capacities of type II presses shall conform to the requirements of table III.

Table III. Sizes and capacities.

Type II

Column or Wall Mounted, Mechanical Presses

Size	Rated tonnage	Largest diameter workpiece accommodated, inches	Largest diameter arbor accommodated, inches	Distance ram to table surface, (ram up), inches	Leverage ratio
3	3	14	2	11-1/4	48:1

Requirements are not less than those shown.

3.4.4 Type III, class 1 presses.

3.4.4.1 Frame. The type III, class 1 press shall have a "C" or gap type frame and be designed for floor mounting. The frame shall support one or more worktables, a ram and ram operating mechanism. The ram shall have sufficient accuracy to maintain the ram axis perpendicular to the table surface within 0.010 inch per 6 inches of ram length. The type III, class 1, style A press shall conform to 3.4.4.2.1 and the type III, class 1, style B press shall conform to 3.4.4.2.2.

3.4.4.2 Worktable.

3.4.4.2.1 Style A. The type III, class 1, style A press shall have a worktable which can be elevated or lowered on a way or slide machined into the face or column of the press frame. The slide shall have a gear rack to allow the table to be raised or lowered by means of a hand cranked pinion mechanism. The press shall have a mechanism that securely clamps the table to the column at any position within the table movement range. The worktable working surface shall be machined and shall have provisions for mounting a removable work support plate conforming to 3.4.4.4, and a slot that will allow

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passage of an arbor of the diameter specified in table IV. The distance from the ram to the worktable surface, with the ram raised and table lowered, shall conform to the requirements of table IV.

3.4.4.2.2 Style B. The type III, class1, style B press shall have sturdy pivots to support two swinging tables, one above the other and allow both tables, individually, to swing into and out of normal operating position. Means shall be provided for locking the worktables in the normal operating position. The upper table surface, lower table surface and base of the press shall be usable to support the workpiece during arbor pressing operations. Each table and the base shall have a machined working surface and provisions for mounting a removable work support plate conforming to 3.4.4.3. Each swing worktable and the base shall have a slot aligned with the ram axis that will allow passage of an arbor of the diameter specified in table IV. The distance from the ram to the worktable work surfaces and base work surfaces shall conform to the requirements of table IV.

3.4.4.3 Work support plate. Type III, class 1, style A and style B presses shall have a work support plate conforming to all requirements of 3.4.1.2 (substituting "table IV" wherever "table I" is mentioned in 3.4.1.2).

3.4.4.4 Ram and ram operating mechanism. All type III, class 1, style A and style B presses shall have a ram and ram operating mechanism conforming to all requirements of 3.4.1.3 (substituting "table IV" wherever "table I" is mentioned in 3.4.1.3).

3.4.4.5 Arbor catcher (type III, class 2, style A presses only). Type III, class 2, style A presses shall have a device that catches the falling arbor as it is pressed out of the workpiece. The device shall consist essentially of a cylindrical or box shaped container lined with a soft metal, such as babbitt, copper, or other suitable metal alloy. The device is not required to have moving parts since it is intended only to "arrest-the-fall" of heavy arbors as they are pressed out of the workpiece, preventing damage to the arbor or the machine base. The device shall be of adequate size to accommodate arbors of the largest diameter specified in table IV and shall be securely mounted on the machine base in alignment with the ram axis.

3.4.4.6 Sizes and capacities. The sizes and capacities of type III, class 1, presses shall conform to the requirements of table IV.

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TABLE IV

SIZES AND CAPACITIES

TYPE III - CLASS 1

FLOOR MOUNTED, MECHANICAL PRESSES

STYLE A - ELEVATING TABLE:

Size	Rated tonnage	Largest diameter workpiece accommodated, inches	Largest diameter arbor accommodated, inches	Distance ram to table surface (ram up), (inches)	Leverage ratio
5	5	20	4-1/2	26	55:1
7	7	27	4-1/2	32	90:1
12	12	20	4-1/2	26	112:1
15	15	27	4-1/2	32	200:1

STYLE B - SWINGING TABLE:

Size	Rated tonnage	Largest diameter workpiece accommodated, inches	Largest diameter arbor accommodated, inches	Distances, ram to work support plates and base work surface (ram up),			Leverage Ratio
				Top table plate	Lower table plate	Base	
15	15	24-1/2	5-1/2	16	30	46	150:1
25	25	37	5-1/2	16-1/2	31-1/2	47	220:1

Requirements are not less than those shown.

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3.4.5 Type III, class 2 presses.

3.4.5.1 Frame. The type III, class 2 press shall have an "H" type frame. The frame shall be fabricated of structural sections such as I-beams, channels, angles, bars, plates, or any combination thereof. The side members shall be ridigly joined together by three or more crossmembers. One crossmember shall serve as a headrail that supports the hydraulic workhead. The crossmember directly below the headrail shall serve as the worktable, and one or more additional crossmembers shall provide additional rigidity to the frame and maintain parallelism of the side members as the table or headrail, as applicable, is raised or lowered. The frame shall have feet of adequate length and strength to stabilize the frame structure and prevent it from tilting or tipping during press operation. Means shall be provided for bolting the frame to the floor. In lieu of the crossmember type table described above, type III, class 2, style B presses shall have a track that supports a roller mounted table that can be rolled out of normal operating position to facilitate loading of heavy workpieces onto the table by means of an overhead crane or hoist. The frame structure, table structure, table support mechanism, and workhead of all type III, class 2 presses shall have sufficient accuracy and rigidity to maintain the ram axis perpendicular to the working surface of the table to an accuracy of 0.050 inch per foot.

3.4.5.1.1 Frame siderails. The siderails shall be either of the open or closed type. The siderails shall have a series of equally spaced holes that allow either the headrail or worktable, or both, to be raised or lowered. The hole spacing shall provide a maximum distance between ram nose and the worktable in accordance with the requirements of table V. The siderails shall support a manually operated, self-locking winch, and necessary cable, chain and pulleys for raising and lowering the headrail or table, as applicable. A set of hardened steel pins shall be provided for supporting the headrail or worktable to the siderails. The pins shall be shaped for easy insertion and removal from the holes. The total shear strength of the pin set shall be not less than double the tonnage rating of the press, The table and table support pins shall be capable of withstanding the bending stresses encountered when operating at 110% of the rated press tonnage without permanent deformation of the table or table support pins. The pin hole spacing in all members shall be to the precision necessary to insure that all pins support equal loads when the workhead is centered on the headrail. The width of the press between the frame siderails shall conform to the requirements of table V.

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3.4.5.1.2 Worktable. The worktable shall be designed to directly support the workpiece, or to support workholding devices (see 3.4.5.5) that support the workpiece. The width between table channels on style A presses shall conform to the requirements for style A presses in table V. The width between table channels on style B presses shall be adjustable over the range specified for style B machines in table V. The worktable of style A machines shall be of the type that is permanently located below the ram axis under all conditions of operation. The worktable for style B presses shall be mounted on rollers that roll in a track mounted on the press frame. The rolling worktable shall be easily rolled into and out of normal operating position below the ram axis so that large workpieces can be readily loaded onto the table in the normal operating position. The table travel distance shall be such that when the table channels are adjusted to the spacing specified in table V, style B, and when the table is in the "loading" position, all table channel edges shall be beyond a perpendicular dropped from the side of the headrail structure (the side of the headrail closest to the extended worktable).

3.4.5.2 Hydraulic workhead. Unless otherwise specified, the hydraulic workhead shall be either the fixed or rolling type. When specified (see 6.2.1), the workhead shall be of the rolling type. Fixed workheads shall be permanently and securely attached to the center of the headrail. Rolling type workheads shall have means for secure clamping at any adjusted position within the range of movement. Presses having a rolling type workhead shall be capable of performing full rated tonnage pressing operations with the workhead at any position within its range of travel without deformation of any component or structural member. The workhead shall contain the hydraulic cylinder, ram piston, ram and ram extension mechanism. The hydraulic pump may be mounted directly on the workhead, or may be mounted at any convenient operating position on the press frame.

3.4.5.3 Hydraulic system. The hydraulic system shall consist essentially of a fluid reservoir, hydraulic pump, cylinder and ram assembly, pressure release valve, pressure gauge and necessary piping, hoses and fittings. The hydraulic system design shall enable the press to deliver the rated tonnage, specified in table V, with not more than 110 pounds of force applied to the end of the pump lever. All components in the hydraulic system shall have a safe working pressure rating not less than 120% of the pressure that can be exerted

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by the pump. Means shall be provided to prevent loading the press beyond 110% of its rated tonnage. Means shall also be provided to prevent extension of the ram beyond its normal "end-of-stroke" position. All components in the system shall be sealed to prevent fluid leakage when the system is operating at full tonnage capacity.

3.4.5.3.1 Fluid reservoir. The fluid reservoir shall have adequate capacity to hold not less than 1-1/4 times the volume of fluid required to fill the system and force the ram to the bottom of its stroke.

3.4.5.3.2 Hydraulic pump, cylinder, and ram. Unless otherwise specified, the machine shall have any one of the types of pump, cylinder and ram mechanism described in 3.4.5.3.2.1 thru 3.4.5.3.2.3 at the option of the supplier. When specified (see 6.2.1), the press shall have the type of pump, cylinder and ram mechanisms selected from 3.4.5.3.2.1 thru 3.4.5.3.2.3 as required by the procuring activity. Regardless of the system furnished, the pump shall have a lever designed to insure comfortable hand operation and prevent lever distortion when the press is developing maximum rated capacity. The pump, or other part of the hydraulic circuit, shall have a pressure release valve that releases pressure back to the reservoir allowing the ram to be returned to the "top-of-stroke" position.

3.4.5.3.2.1 Dual-stage pump type system. This type system shall consist of a pump that has high volume-low pressure stage and a low volume-high pressure stage. The high volume-low pressure stage shall provide rapid ram extension to the arbor prior to the arbor pressing operation, and, when used with a double acting ram cylinder having no spring return mechanism, shall provide rapid return of the ram to the "top-of-stroke" position. The low volume-high pressure stage shall be capable of delivering adequate pressure to enable the press to exert the rated tonnage specified in table V, without requiring more than 110 pounds of force on the end of the pump operating lever.

3.4.5.3.2.2 Single-stage pump with rack and pinion ram extension mechanism. This type system shall consist of a single pump capable of developing adequate pressure to enable the press to exert the tonnage specified in table V, without requiring more than 110 pounds of force on the end of the pump operating lever. The ram shall have a mechanical system, such as a handwheel operated rack and pinion mechanism, that provides rapid ram extension to the arbor prior

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to the arbor pressing operation and returns the ram to the "top-of-stroke" position after completion of the pressing operation and after actuation of the pressure release valve.

3.4.5.3.2.3 Single-stage pump with threaded ram extension mechanism. This type system shall consist of a single-stage pump capable of developing adequate pressure to enable the press to exert rated tonnage specified in table V, without requiring more than 110 pounds of force on the end of the pump operating lever. The ram shall have threaded extension for rapidly extending the ram to contact the arbor prior to the actual arbor pressing operation.

3.4.5.4 Auxiliary 3 ton, rapid ram movement capabilities. Unless otherwise specified (see 6.2.1, all type III, class 2 presses are required to have means of exerting a ram force of 3 tons, or more, without requiring a large number of pump strokes. This may be accomplished by:

(a) Utilizing a dual speed pump system (see 3.4.5.3.2.1) capable of providing a 3 ton, or more, ram force at a ram speed of .25-inch or more per stroke of the pump, with a force on the pump lever not greater than 110 pounds.

(b) Utilizing a single-stage pump with a rack and pinion type ram extension device (see 3.4.5.3.2.2) capable of mechanically exerting a 3-ton force at a leverage ratio not less than 48 to 1. If the pump system does not meet the requirements of (a) or (b) above, an auxiliary 3-ton mechanical press meeting or exceeding the requirements of type I, class 1, size 3B, or type II, size 3 of this specification shall be furnished. The 3-ton auxiliary mechanical press, when provided, shall be securely mounted at a convenient location on the press frame.

3.4.5.5 Work support device. All type III, class 2 presses shall have two work support plates that rest on the table channels and support the workpiece. The work support plates shall be movable along the entire width of the table to provide the necessary support for any workpiece within the size range of the machine. The upper surface of the plates shall be flat and the bottom surface shall have a tongue, step, shoulder, or other means that holds the plate in alignment, prevents the plate from sliding off the table, but allow free sliding motion on the table surface. In addition, a pair of

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V-block type work support plates, suitable for supporting a round bar during straightening operations shall be furnished with each press. If the supplier normally furnished a single pair of invertable blocks suitable for supporting flat surfaces using one side, and supporting round bars using the other side, the pair of invertable blocks may be supplied in lieu of the two pairs of supports required above.

3.4.5.6 Pressure gauge. All hydraulic presses shall have a pressure gauge, calibrated in tons of ram force. The gauge shall be legible from the operators normal work station and shall not be obstructed by any part of the press.

3.4.5.7 Sizes and capacities. The sizes and capacities of type III, class 2 presses shall conform to the requirements of table V.

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TABLE V
SIZES AND CAPACITIES

TYPE III - CLASS 2
FLOOR MOUNTED, HYDRAULIC PRESSES

STYLE A - STANDARD (NONROLLING) TABLE						
Size	Rated tonnage	Width between frame siderails, inches	Width between table channels (max. arbor size), inches	Distance, ram nose to table, inches	Vertical travel of table or headrail, inches	Ram piston travel inches
17	17-1/2	20	7	34	35	2
20	20	24	5	37	34	6
25	25	24-1/4	5-3/16	39	35	5
30	30	32	7-1/2	35	35	2-1/2
40	40	30-1/2	7	43	36	7
50	50	31	7	40	35	4
60	60	36	8	40	36	7
75	75	42	8	36	30	4-1/2
80	80	48	8	40	32	7
100	100	45-1/2	10	36	30	14
125	125	48	9-1/2	25	20	8
150	150	48	10	25	20	8
200	200	48	18	30	24	14

STYLE B - ROLLER MOUNTED TABLE							
Size	Rated tonnage	Width between upright columns, inches	Width between table channels adjustable over a range of, inches	*Table channel spacing for unobstructed table loading, inches	Distance, ram nose to table, inches	Vertical travel of table or headrail inches	Ram piston travel, inches
50	50	42	0 to 30	8-1/2	40-1/2	28	12
75	75	45	0 to 29	10-1/2	39	28	11
100	100	48	0 to 28	11	60	44	14
150	150	48	0 to 27	11-3/4	58	44	14

* These dimensions represent the widest table channel spacing that will allow the table to be rolled to a position in which all table channel edges are beyond a perpendicular dropped from the headrail structure (see 3.4.5.1.2).

Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure, or less, to the larger figure, or more.

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3.5 Performance requirements. All presses covered by this specification shall be capable of performing any function for which they are designed, within their rated tonnage capacity. The presses covered herein shall be capable of exerting a force equal to 110 percent of their maximum rated tonnage capacity without permanent distortion or other damage to any component.

3.6 Standard equipment. Unless otherwise specified (see 6.2.1), the presses shall be furnished with all standard equipment normally furnished by the manufacturer including the following:

(a) One set of tools normally furnished with each press

(b) Two ram noses, one flat, and one vee

3.7 Optional equipment. Optional equipment furnished shall be as specified (see 6.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.2, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all test in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Each press shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

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4.4 Examination. The press and associated equipment shall be examined to determine compliance with the requirements in 3.3 through 3.7.

4.5 Tests.

4.5.1 Operational test. The press shall be operated at no load for a period of time sufficient to assure proper operation of the press, all controls, adjusting mechanisms, and accessories.

4.5.2 Performance test. Each press shall be loaded and tested for proper operation when loaded to 110 percent of maximum rated capacity. Type I, class 2 and type III, class 2 presses shall retain the load for a period of not less than one hour. Either during or after testing, the press shall be examined for permanent distortion, breakage, hydraulic leaks, proper operation of the pressure gauge and ram retraction mechanism, and any other evidence of nonconformance to this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with MIL-M-18058. Level of preservation and packing shall be as specified (see 6.2.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to the marking required by the contract or order (when specified see 6.2.1) marking shall be in accordance with Federal Standard No. 123.

6. NOTES

6.1 Intended use. The presses covered by this specification are for use in machine, production, and repair shops at shore facilities and on board ships.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

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a. Title, number and date of this specification.
b. Type, class, style and size required (see 1.2.1).
c. First article approval, if required (see 3.2).
d. Specify exceptions and additional safety and health requirements (see 3.3.3 and 6.5).
e. Painting, if different (see 3.3.11).
f. Specify fungus proofing, when required (see 3.3.15).
g. Nameplate, if different (see 3.3.16).
h. Lubrication chart, if not required (see 3.3.17).
i. Specify pedestal or floor stand, when required (see 3.4.1.1 and 3.4.2.1).
j. "C" or gap frame design, when required (see 3.4.2).
k. Specify rolling type workhead, when required (see 3.4.5.2).
l. Specify type of pump, cylinder, and ram mechanism, when a specific type is required (see 3.4.5.3.2).
m. When high speed 3-ton pressing capability is not required (see 3.4.5.4).
n. Standard equipment required, when different (see 3.6).
o. Optional equipment if required; fully describe optional equipment required (see 3.7).
p. First article inspection, if different (see 4.2).
q. Quality conformance inspection, if different (see 4.3).
r. Preservation, packing and marking if different and level of preservation and packing required (see 5.1).
s. Civil agency special marking, if required (see 5.2).

6.3 Contract data requirements. For military procurements, required technical data, such as operator manuals, parts lists, foundation and anchor bolt plans, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. A first article comprises a preproduction item or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, test and approval of the First Article.

6.5 Safety and health environmental determinations. In order that equipment integrated into the user's operational environment will comply with the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission,

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noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

6.6 Mercury prohibited. When the use of mercury or its compounds in the machine and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

MILITARY CUSTODIANS:
Army - WC
Navy - SH
Air Force - 99

Review activities:
Army - WC
Navy - YD, SH
Air Force - 99
DSA - GS

User activities:
Army - AV, SM, WV, ME
Navy - AS, MC

Preparing activity:
DSA - IP

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Project Number: 3444-0015

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 50 cents each.

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DOCUMENT IDENTIFIER AND TITLE

OO-P-636D - Presses, Arbor, Hand-Operated

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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